environments. Also, the master can have several jobs running on different nodes simultaneously, which not only saves time but also improves the efficiency.

Best practices

A best practice is a technique or process that is generally considered as surpassing any other method of doing something, as it produces the best possible outcomes in known scenarios. Best practices are followed in every field of work to meet benchmarks and quality standards. Jenkins best practices result in a better sense of how Jenkins plays a crucial role throughout the project life cycle for software developers and senior management. Some of the best practices that should be followed while using Jenkins are listed in Table 1.

In the next section, we will understand the basics of containers.

Docker is an open platform to develop, deploy and run applications within containers. The use of containers allows the developer to build, test and deploy faster in an isolated environment and separate the application from the host's environment.

Containers

People often get confused between containers and virtual machines (VMs). Both have the same objective -- to isolate the application from the host machine and provide its dependencies in an independent module. When Linux containers are used to deploy an application, it is known as containerisation, which is gaining popularity for the reasons listed in Table 2.

Now, as we have a basic idea about Docker and containers, we shall explore the underlying abstraction of Docker, bit by bit.

Docker registry

When a Docker container is completely designed and is stable to run, this container image can be published to the registry. The registry can be public or private -- Docker provides its own registry known as Docker Hub. This enables the Docker hosts to pull the image from the registry and run it on their system. As discussed, the Docker host only needs the Docker image's name and its tag.

Jenkins on Docker

Table 4: Docker commands

Command	Description
-u or –user	When a container is running, this specifies which user the environment is logged on to. The jenkinsci/blueocean image by default logs in as user jenkins. But we have logged in as a root user to tackle the permission related hurdles.
-d	This signifies that when the container is running, it should be detached from the current terminal and run in the background.
-p or –publish	This is used to publish the ports from inside the container to the outer environment. The ports inside the container and those of the host are totally independent. Thus, if we need to communicate to the container via ports, we need to bridge it with some external port. Here, 8080 is used, because Jenkins by default runs on Port 8080; Port 50000 is used for JNLP connections between master and agent.
-v	Jenkins generates its data into the Jenkins Home directory. But the moment a container is deleted, all that data will be lost. In order to retain the data, we can use either Docker volumes or mount the host's directory to the Docker container. Here, we have tied the host's directory to the Docker container. We have mapped Jenkins Home to store Jenkins data and the Docker directory, in order to access Docker services inside the Docker container.
--name	This is used to give a name to the container. If not given, Docker will give a random name.
--restart always	This is used to make sure if anything happens to the Docker daemon and it restarts, or if a container stops, this property will restart the container again automatically.

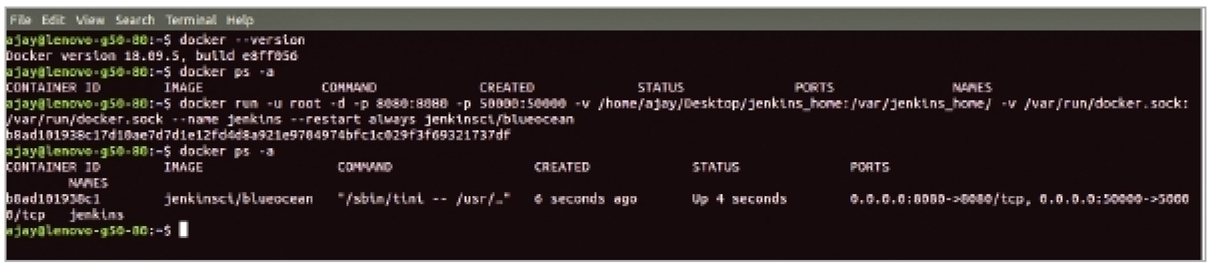

Figure 5: Container details

Figure 6: Jenkins home page